

DDSA vs deterministic-ODE fitting of aggregated incidence counts

DDSA paper

Overview

This notebook contrasts discrete dynamic survival analysis (DDSA) with fitting a deterministic compartmental ODE to aggregated daily counts under a Poisson or Negative-Binomial observation model (see the [epirecipes notebook](#)).

Epidemiological data can be presented as a line list of onset and recovery events. A common path aggregates those events into a daily incidence curve and fits a continuous-time ODE, evaluating the solution at integer days. DDSA is built in discrete time and uses the line list directly.

To compare the two approaches, we generate a synthetic outbreak with a known non-exponential infectious period (Erlang-4; $k = 4$) and fit five models. The table below varies the forward model, how the infectious-period (IP) shape is treated (*assumed* vs *inferred*), whether the mean IP (removal rate γ) is inferred or fixed, and the observation likelihood:

Model	Forward model	IP shape	Mean IP / γ	Inference likelihood
1. ODE + Poisson	Continuous SIR ODE	Exponential (single I), <i>assumed</i>	γ <i>inferred</i>	Poisson on daily counts
2. ODE + NegBin	Continuous SIR ODE	Exponential (single I), <i>assumed</i>	γ <i>inferred</i>	Negative-Binomial on daily counts
3. ODE (staged) + NegBin	Continuous SIR ODE, 4 I -stages	Erlang-4 (four staged I), <i>assumed</i>	γ <i>inferred</i>	Negative-Binomial on daily counts
4. ODE, γ fixed	Continuous SIR ODE	Exponential (single I), <i>assumed</i>	γ <i>fixed</i> from literature ($\mu = 4$ d)	Negative-Binomial on daily counts

Model	Forward model	IP shape	Mean IP / γ	Inference likelihood
5. DDSA	Discrete GLCT map	Discrete phase-type (NegBin), <i>inferred</i> (shape r)	mean IP <i>inferred</i>	Multinomial onsets, phase-type recovery intervals, Binomial final size

The four ODE comparators represent realistic practitioner choices when only an incidence curve is available. Models 1 and 2 are the default workflow: fit a standard exponential SIR ODE to daily counts, with Poisson or Negative-Binomial noise. Model 3 represents a practitioner who suspects a non-exponential IP (for example, from qualitative knowledge that the infectious period is less variable than the exponential) and uses a staged ODE to reflect this, but still leaves γ free because no reliable external estimate exists. Model 4 represents one approach of anchoring the IP mean from prior literature — fixing γ at a published estimate — while retaining the default exponential shape. DDSA (Model 5) represents the alternative when individual-level records are available: the line list supplies recovery intervals that make joint inference of IP shape and mean possible, without external assumptions.

This comparison illustrates specific instances rather than suggesting a best practice recipe. ODE models can use any integrator or count observation model. DDSA is similarly modular, the recovery term can be any discrete phase-type distribution (geometric, NegBin/DPH, boxcar) with shape fixed or inferred, and onset or final-size terms can be dropped when those records are unavailable.

Setup

```
using Turing, Distributions, Random, Statistics, LinearAlgebra, DataFrames
using OrdinaryDiffEq, Plots, StatsPlots
include("../PhaseTypeDistributions.jl")
Random.seed!(1234)
```

DDSA model and helper functions

This section defines the DDSA components: helper functions, the discrete Generalised Linear Chain Trick (GLCT) map, a synthetic line-list generator, and the full DDSA model.

```

"""Normalise a non-negative vector to sum to 1 (uniform fallback if all-zero)."""
function safe_normalise(f::AbstractVector{T}, n::Int) where T
    f_pos = ifelse(isnan.(f) .| .!isfinite.(f), zero(T), max.(f, zero(T)))
    sF = sum(f_pos)
    (sF > T(1e-15) && isfinite(sF)) ? f_pos ./ sF : fill(one(T)/n, n)
end

"""Convert a rate to a per-interval proportion, `1 - exp(-r*t)`. """
@inline function rate_to_proportion(r, t=1.0)
    (r < 0 || isnan(r)) && return 0.0
    isinf(r) && return 1.0
    res = 1 - exp(-r * t)
    isnan(res) ? 0.0 : max(min(res, 1.0), 0.0)
end

"""Run the discrete GLCT forward map over `nsteps` steps from initial infected
fraction ``, with phase-type generator `(Tm, )`, transmission ``, and step `Δt`.
Returns `(S, , f)`: the susceptible-fraction history, the attack fraction ``, and
the normalised infection-time PMF `f`. """
function simulate_discrete_glct(Tm, , , nsteps::Int, Δt)
    S = 1.0 - ; x = .*
    S_hist = Any[S]
    for _ in 1:nsteps
        new_inf = rate_to_proportion( * sum(x), Δt) * S
        x = Tm' * x + .* new_inf
        S -= new_inf
        push!(S_hist, S)
    end
    = S_hist[1] - S_hist[end]
    f = max.(-diff(S_hist), 0.0)
    sF = sum(f); f = sF > 0 ? f ./ sF : fill(one(eltype(f))/length(f), length(f))
    return (S=S_hist, =, f=f)
end

"""Simulate one synthetic line list from the discrete GLCT truth with transmission ``,
phase-type IP generator `(Tm, )`, initial infected fraction ``, and population `N`,
over `n_events` steps of size `Δt`. Draws the final size `K ~ Binomial(N, )`, the onset
histogram `th ~ Multinomial(K, f)`, and `M + K` recovery intervals `delta`. Returns a
`Dict` with keys `:K, :th, :delta, :, :f`. """
function generate_discrete_glct_data(, Tm, , , N, M, n_events, Δt)
    res = simulate_discrete_glct(Tm, , , , n_events, Δt)
    K = rand(Binomial(N, res.))

```

```

th = K > 0 ? rand(Multinomial(K, res.f ./ sum(res.f))) : zeros(Int, n_events)
delta = Int[]
dph = PhaseTypeDistributions.DPH(, Tm)
for _ in 1:(M + K)
    u = rand(); cum = 0.0; rt = 1
    while cum < u
        cum += PhaseTypeDistributions.pmf(dph, rt)
        cum >= u && break
        rt += 1
    end
    push!(delta, rt)
end
return Dict{:K => K, :th => th, :delta => delta, : => res., :f => res.f}
end

_prior_hi = 2.5 # shared -prior upper bound for all 5 models

"""DDSA model: final size (Binomial), onset times (Multinomial), and recovery intervals (Negl
@model function dph_full(N::Int, M::Int, nsteps::Int, K::Int,
                        th::Vector{Int}, delta::Vector{Int}, nb_r_max::Int, Δt::Float64)
    ~ Uniform(1e-4, _prior_hi)
    ~ Beta(1.0, N - 1)
nb_p ~ Beta(1.0, 1.0)
nb_r ~ DiscreteUniform(1, nb_r_max)
dph = PhaseTypeDistributions.dph_negative_binomial(nb_r, nb_p)
result = simulate_discrete_glct(dph.S, dph., , , nsteps, Δt)
_valid = clamp(result., 0.001, 0.999)
f_safe = safe_normalise(result.f, nsteps)
K ~ Binomial(N, _valid)
if K > 0
    if all(x -> isfinite(x) && !isnan(x), f_safe) && abs(sum(f_safe) - 1) < 0.01
        th ~ Multinomial(K, f_safe)
    else
        Turing.@addlogprob! -Inf
    end
end
end
for i in 1:(M + K)
    delta[i] ~ dph
end
end
end

```

Synthetic line-list data

We simulate an Erlang-4 outbreak with $\beta = 0.5/\text{step}$, $N = 99$, $\rho = 1/N$ (one index case), and a timeline length of $T = 40$ d (continuous Erlang mean $\mu = 4$ d). At $\Delta t = 1$ d the discrete dwell-time mean is $E[W] \approx 6.33$ d, giving discrete $R_0 = \beta E[W] \approx 3.16$. The continuous calibration $R_0 = \beta\mu = 2.0$ differs because $\Delta t = 1$ inflates the effective mean IP.

```
N_pop    = 99
R0_cal, _ip, k, Δt = 2.0, 4.0, 4, 1.0
_true    = 1.0 / N_pop
nsteps   = 40
_true    = (R0_cal / _ip) * Δt
dph_nom  = PhaseTypeDistributions.dph_erlang_from_mean(k, _ip, Δt)
EW       = PhaseTypeDistributions.mean(dph_nom)
R0_imp   = _true * EW

data = generate_discrete_glct_data(_true, dph_nom.S, dph_nom. , _true, N_pop, 1, nsteps, Δt)
K, th, delta = data[:K], data[:th], data[:delta]
N = N_pop
incidence = th
```

Truth: $\beta=0.5$, $E[W]=6.33$ d, $R_{0_imp}=3.16$, $R_{0_cont}=2.0$

Obs: $K=91$, $\text{attack}=0.919$, $\text{recovery mean}=6.41$ d, $CV=0.246$

Count-based comparators (continuous ODE + Poisson/NegBin counts)

Models 1–4 follow the `ode_turing` recipe: solve a continuous-time SIR ODE, read expected daily new infections from cumulative incidence, and attach a Poisson or Negative-Binomial count likelihood. Each evaluation solves the ODE inside the likelihood loop using an explicit RK integrator (Tsit5) and rejects non-finite solutions.

```
NegBin2(m, ) = NegativeBinomial( , clamp( / ( + m), 1e-6, 1.0 - 1e-6))

function sir_exp_ode!(du, u, p, t) # exponential IP (single I) -- shape mis-specified
    (S, I, R, C) = u; ( , ) = p; Npop = S + I + R
    inf = * I / Npop * S
    du[1] = -inf; du[2] = inf - *I; du[3] = *I; du[4] = inf
end
```

```

function sir_erlang_ode!(du, u, p, t)      # Erlang-4 IP (4 staged I) -- correctly staged
    S = u[1]; I = @view u[2:5]; (, ) = p
    Npop = S + sum(I) + u[7]
    inf = * sum(I) / Npop * S
    k = k *
    du[1] = -inf
    du[2] = inf      - k * I[1]
    du[3] = k * I[1] - k * I[2]
    du[4] = k * I[2] - k * I[3]
    du[5] = k * I[3] - k * I[4]
    du[7] = k * I[4]      # R
    du[6] = inf          # cumulative incidence C
end

# (1) exponential IP, gamma free, Poisson counts -- "sample a Poisson" practice.
@model function count_exp_pois(y, N)
    l = length(y)
    i0 ~ Uniform(1e-4, 0.2); ~ Uniform(1e-4, _prior_hi); ~ Uniform(0.02, 1.0)
    IO = i0 * N
    prob = ODEProblem(sir_exp_ode!, [N - IO, IO, 0.0, 0.0], (0.0, float(l)), [, ])
    sol = solve(prob, Tsit5(), saveat = 1.0)
    C = Array(sol)[4, :]; newI = clamp.(diff(C), 1e-6, 1e7)
    (length(newI) == l && all(isfinite, newI)) || (Turing.@addlogprob! -Inf; return)
    for i in 1:l; y[i] ~ Poisson(newI[i]); end
end

# (2) exponential IP, gamma free, NegBin counts -- same, with overdispersion.
@model function count_exp(y, N)
    l = length(y)
    i0 ~ Uniform(1e-4, 0.2); ~ Uniform(1e-4, _prior_hi)
    ~ Uniform(0.02, 1.0); ~ Exponential(5.0)
    IO = i0 * N
    prob = ODEProblem(sir_exp_ode!, [N - IO, IO, 0.0, 0.0], (0.0, float(l)), [, ])
    sol = solve(prob, Tsit5(), saveat = 1.0)
    C = Array(sol)[4, :]; newI = clamp.(diff(C), 1e-6, 1e7)
    (length(newI) == l && all(isfinite, newI)) || (Turing.@addlogprob! -Inf; return)
    for i in 1:l; y[i] ~ NegBin2(newI[i], ); end
end

# (3) Erlang-4 IP, gamma free, NegBin -- correct shape, but assumed not learned.
@model function count_erlang(y, N)
    l = length(y)

```

```

i0 ~ Uniform(1e-4, 0.2); ~ Uniform(1e-4, _prior_hi)
~ Uniform(0.02, 1.0); ~ Exponential(5.0)
IO = i0 * N
u0 = [N - IO, IO, 0.0, 0.0, 0.0, 0.0, 0.0]
prob = ODEProblem(sir_erlang_ode!, u0, (0.0, float(1)), [, ])
sol = solve(prob, Tsit5(), saveat = 1.0)
C = Array(sol)[6, :]; newI = clamp.(diff(C), 1e-6, 1e7)
(length(newI) == 1 && all(isfinite, newI)) || (Turing.@addlogprob! -Inf; return)
for i in 1:l; y[i] ~ NegBin2(newI[i], ); end
end

# (4) exponential IP, gamma fixed at the continuous-time mean IP from prior literature (_ip
# a practitioner reads "mean IP 4 days" in prior studies and sets accordingly.
_fix = 1.0 / _ip
@model function count_exp_fixed(y, N, f)
l = length(y)
i0 ~ Uniform(1e-4, 0.2); ~ Uniform(1e-4, _prior_hi); ~ Exponential(5.0)
IO = i0 * N
prob = ODEProblem(sir_exp_ode!, [N - IO, IO, 0.0, 0.0], (0.0, float(1)), [, f])
sol = solve(prob, Tsit5(), saveat = 1.0)
C = Array(sol)[4, :]; newI = clamp.(diff(C), 1e-6, 1e7)
(length(newI) == 1 && all(isfinite, newI)) || (Turing.@addlogprob! -Inf; return)
for i in 1:l; y[i] ~ NegBin2(newI[i], ); end
end

```

Infection Timing Equivalence

Independent Poisson daily counts, conditioned on their total K , follow a Multinomial(K, f) distribution—identical to the infection-time term in DDSA. The count–Poisson log-likelihood therefore factorises as

$$\sum_t \log \text{Pois}(y_t; \mu_t) = \underbrace{\log \text{Multinomial}(y; K, f)}_{\text{DDSA infection-time term}} + \underbrace{\log \text{Pois}\left(K; \sum_t \mu_t\right)}_{\text{final-size term}}, \quad f_t = \mu_t / \sum_s \mu_s.$$

DDSA uses this multinomial timing term but replaces the Poisson final-size factor with Binomial(N, τ) to explicitly account for a known at-risk population N . It also introduces a recovery-interval term that aggregated counts cannot capture. Assuming Poisson noise, a correctly specified infectious period, and large N , both approaches yield consistent infection timing. They diverge only when the line list provides higher-resolution data: individual recovery

durations (which identify the IP mean and shape) and tighter small- N final-size constraints. For the ODE model, μ_t is sampled from a continuous solution at integer days, so its timing PMF f matches DDSA's only up to discretisation error. The following Julia implementation demonstrates this factorisation on the exponential ODE's mean curve:

```
prob0 = ODEProblem(sir_exp_ode!, [N - _true*N, _true*N, 0.0, 0.0],
                  (0.0, float(nsteps)), [_true, _fix])
C0     = Array(solve(prob0, Tsit5(), saveat = 1.0))[4, :]
_ode   = max.(diff(C0), 1e-9)                # expected daily new infections
f_ode  = _ode ./ sum(_ode)
ll_pois = sum(logpdf.(Poisson.(_ode), incidence)) # count + Poisson, all days
ll_multi = logpdf(Multinomial(K, f_ode), incidence) # = DDSA's infection-time term
ll_total = logpdf(Poisson(sum(_ode)), K)           # Poisson on the grand total
```

```
count-Poisson    Σ logPois(y ; )    = -67.4417
Multinomial(y;K,f) [timing term]    = -63.1071
Poisson(K; total) [final size]     = -4.3346
timing + final size                = -67.4417
identity holds: true
```

Fitting all models

All five models are sampled with 2000 warm-up and 2000 posterior draws. The three γ -free count models use NUTS with target acceptance 0.95 and `max_depth` 12. The γ -fixed model uses standard NUTS (target acceptance 0.8). DDSA uses a Gibbs scheme: Metropolis-Hastings for the discrete shape r and NUTS on (β, ρ, p) .

```
n_warm, n_post = 2000, 2000
nuts_count = NUTS(n_warm, 0.95; max_depth=12, Δ_max=1000.0)
ch_pois = sample(count_exp_pois(incidence, N), nuts_count, n_post; progress=false)
ch_exp  = sample(count_exp(incidence, N), nuts_count, n_post; progress=false)
ch_erl  = sample(count_erlang(incidence, N), nuts_count, n_post; progress=false)
ch_fix  = sample(count_exp_fixed(incidence, N, _fix), NUTS(1000, 0.8), n_post; progress=false)

nb_r_max = min(10, minimum(delta))
gibbs = Gibbs(:nb_r => MH(), (:, :, :nb_p) => NUTS(n_warm, 0.8; max_depth=10, Δ_max=1000.0))
r_init = Int(clamp(minimum(delta), 1, nb_r_max))
p_init = clamp(r_init / max(mean(delta), r_init + 0.5), 0.05, 0.95)
_obs   = K / N
R0_h   = clamp(-log(1 - min(_obs, 0.999)) / max(_obs, 1e-6), 0.5, 5.0)
_init  = clamp(R0_h * p_init / r_init, 1e-4, _prior_hi)
```



```

init    = (;    = _init,    = 1.0/N, nb_p = p_init, nb_r = r_init)
ch_ddsa = sample(dph_full(N, 1, nsteps, K, th, delta, nb_r_max, Δt), gibbs, n_post;
                 initial_params = init, progress=false)

```

Results

Convergence diagnostics ($\hat{R}$, ESS, divergences) and posterior medians with 95% credible intervals for β , mean IP, and R_0 are reported by model.

```

q(v) = "$(round(median(v),digits=3)) [$ (round(quantile(v,0.025),digits=3)), $(round(quantile(v,0.975),digits=3))]"
qi(v) = "$(Int(round(median(v)))) [$ (Int(round(quantile(v,0.025))))], $(Int(round(quantile(v,0.975))))]"
p, p = vec(ch_pois[:]), vec(ch_pois[:]); R0p = p ./ p; ipp = 1.0 ./ p
e, e = vec(ch_exp[:]), vec(ch_exp[:]); R0e = e ./ e; ipe = 1.0 ./ e
r, r = vec(ch_erl[:]), vec(ch_erl[:]); R0r = r ./ r; ipr = 1.0 ./ r
f = vec(ch_fix[:]); R0f = f ./ _fix
d, pr, rr = vec(ch_ddsa[:]), vec(ch_ddsa[:nb_p]), vec(ch_ddsa[:nb_r])
ipd = rr ./ pr; R0d = d .* ipd

function conv(nm, ch)
    s = DataFrame(summarystats(ch)); pn = propertynames(s)
    rh = :rhat in pn ? filter(isfinite, collect(skipmissing(s.rhat))) : Float64[]
    ecraw = :ess_bulk in pn ? s.ess_bulk : (:ess in pn ? s.ess : Float64[])
    ec = filter(isfinite, collect(skipmissing(ecraw)))
    dv = try Int(sum(ch[:numerical_error])) catch; 0 end
    mr = isempty(rh) ? NaN : maximum(rh); me = isempty(ec) ? NaN : minimum(ec)
    ess_str = isnan(me) ? "NaN" : string(Int(round(me)))
    println(" $nm: max R=$(round(mr,digits=3)), min ESS=$ess_str, divergences=$dv")
end

let pr = rand(Uniform(1e-4, _prior_hi), 200_000), pr = rand(Uniform(0.02, 1.0), 200_000)
global R0_prior = pr ./ pr
r0_ci = round.(quantile(R0_prior,[0.025,0.975]),digits=2)
println("prior R0: $(round(median(R0_prior),digits=2)) [$ (r0_ci[1]), $(r0_ci[2])]" )
end

```

prior R0: 2.45 [0.13, 29.63]

Convergence

ODE-exp Poisson (free): max R=1.0, min ESS=290, divergences=0

```

ODE-exp NegBin ( free): max R=1.007, min ESS=463, divergences=0
ODE-erlang NB  ( free): max R=1.005, min ESS=454, divergences=0
ODE-exp NB     ( fixed): max R=1.0, min ESS=783, divergences=0
DDSA           : max R=1.001, min ESS=383, divergences=0

```

Posterior median [95% CrI]

truth: =0.5, E[W]=6.33 d, R0=3.16

ODE-exp Poisson (free): =0.483 [0.376, 0.899] meanIP=16.541 [2.239, 46.231] R0=7.695 [1.7, 15.1]

ODE-exp NegBin (free): =0.524 [0.378, 1.053] meanIP=12.114 [1.688, 46.63] R0=6.062 [1.7, 15.1]

ODE-erlang NB (free): =0.436 [0.327, 0.654] meanIP=11.123 [2.853, 45.751] R0=4.732 [1.7, 15.1]

ODE-exp NB (fixed): =0.686 [0.592, 0.783] meanIP=4.0 d (fixed) R0=2.742 [2.369, 3.13]

DDSA (line list): =0.49 [0.431, 0.559] meanIP=6.379 [6.028, 6.799] R0=3.127 [2.746, 3.508]

Figure

Panel (A) shows observed daily incidence with posterior-predictive medians and 95% bands for the exponential-NegBin ODE and DDSA. All five models fit the incidence curve similarly. Panels (B) and (C) show posterior densities for R_0 and mean infectious period, with the prior (dotted) and true values (dashed) indicated.

The count models with γ free (blue, purple) return posteriors covering nearly the full prior range. The width is similar whether or not the infectious period is correctly staged, suggesting the incidence curve provides limited joint information on β and γ . Fixing γ (green) concentrates the R_0 posterior but shifts it below the truth: the continuous-time rate $\gamma = 1/4$ overestimates the removal rate relative to the effective discrete infectious period of 6.33 d. DDSA (red), informed by the individual recovery intervals, recovers both R_0 and the mean infectious period without fixing either.

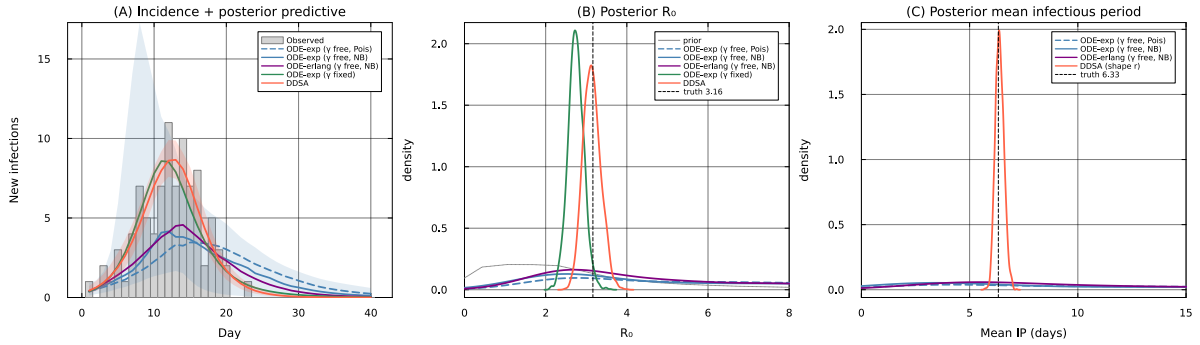


Figure 1

Discussion

Count likelihoods and DDSA share the same infection-timing information: a Poisson count likelihood conditioned on its total is exactly DDSA’s multinomial onset term. All five models fit the incidence curve comparably, so the differences between them lie not in goodness of fit but in what each recovers about the underlying epidemic parameters.

Count models with γ free (Models 1–3) return posteriors covering much of the prior for both R_0 and the mean IP; this holds whether or not the infectious period is correctly staged. Fixing γ from the literature (Model 4) narrows the R_0 posterior but introduces a downward bias: the continuous-time rate $\gamma = 1/4$ overestimates removal relative to the effective discrete IP of 6.33 d, so the inferred $R_0 = \hat{\beta}/\gamma_{\text{fix}}$ undershoots the truth even as $\hat{\beta}$ adjusts upward to fit the incidence curve. This bias was not apparent from incidence data in this example. DDSA (Model 5), using the same incidence together with recovery intervals and final size, recovers R_0 , mean IP, and dwell-time shape without fixing any IP parameter externally.

When only daily incidence is available, count models can be the natural choice. When a line list is also available, the recovery intervals provide information that aggregated counts cannot: they can help identify both the mean IP and its shape, and may reveal biases introduced by external IP assumptions that are difficult to detect from incidence data alone.